Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	Kunal Yadav
Roll No. :	64
Experiment No.:	**4**
Title of the Experiment:	**Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets**
Date of Performance:	28/07/26
Date of Submission:	04/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : DR . TATWADARSHI P N

Signature :

Date :

Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:
    a. One-sample t-test
    b. Two-sample t-test
7. Calculate the test statistic and p-value.
8. Compare the p-value with the selected significance level.
9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ.

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where $x$is the number of observations with a particular characteristic.

Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p - value < \alpha$, reject H_0.
- If $p - value \geq \alpha$, fail to reject H_0.

t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

$$Diabetes\ Outcome and Gender$$

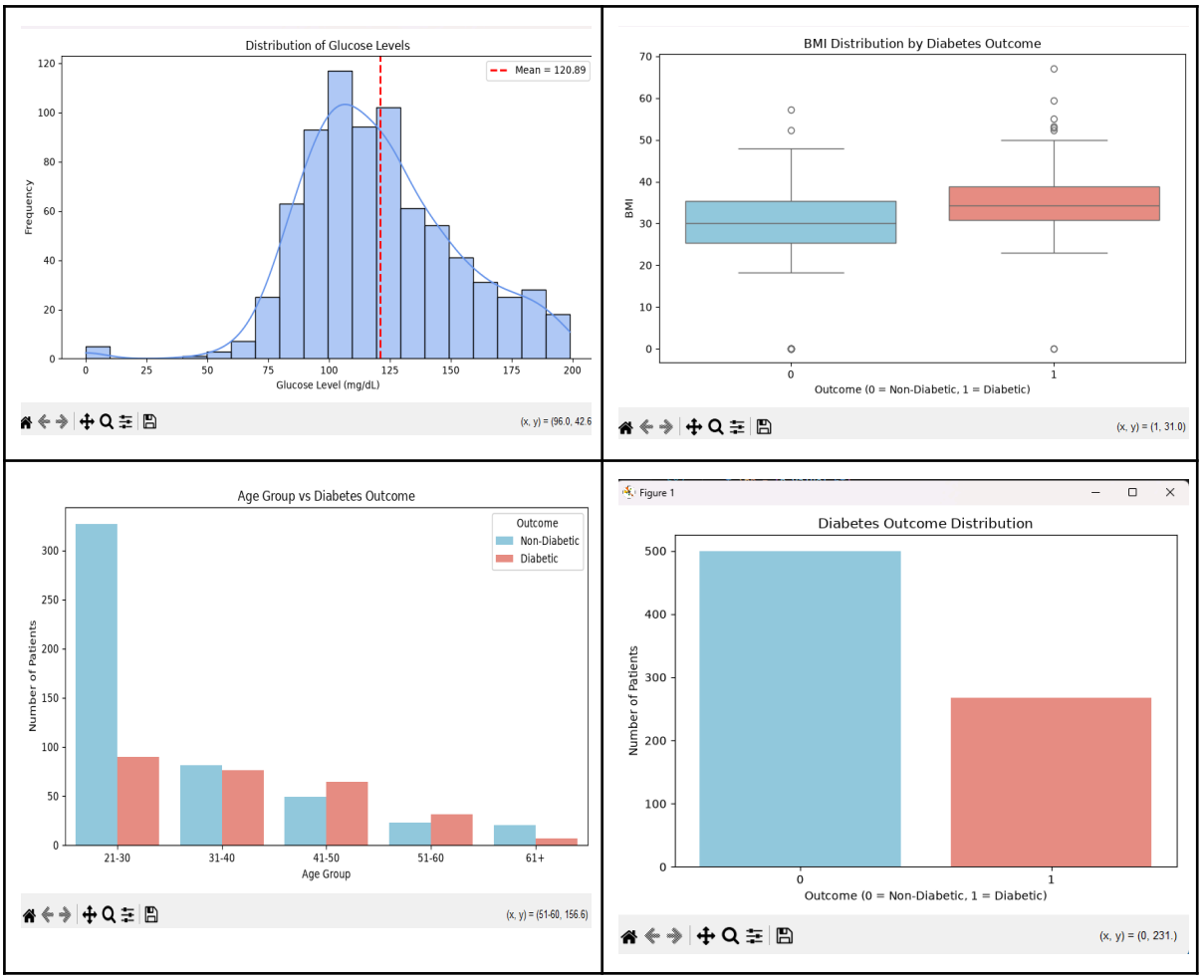
where applicable.

Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
urlhttps://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.cs
cols = ["Pregnancies","Glucose","BloodPressure","SkinThickness",
    "Insulin","BMI","DiabetesPedigreeFunction","Age","Outcome"]
df = pd.read_csv(url, names=cols)
```

```
sns.histplot(df["Glucose"], bins=20, kde=True, color="skyblue")
plt.axvline(df["Glucose"].mean(), color="red", linestyle="--",
            label=f"Mean = {df['Glucose'].mean():.2f}")
plt.title("Distribution of Glucose Levels")
plt.xlabel("Glucose")
plt.ylabel("Frequency")
plt.legend()
plt.show()
sns.boxplot(x="Outcome", y="BMI", data=df,
            hue="Outcome", palette=["skyblue", "salmon"],
            legend=False)

plt.title("BMI Distribution by Diabetes Outcome")
plt.xlabel("Outcome (0 = Non-Diabetic, 1 = Diabetic)")
plt.ylabel("BMI")
plt.show()
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[20,30,40,50,60,100],
    labels=["21-30","31-40","41-50","51-60","61+"]
)
sns.countplot(x="AgeGroup", hue="Outcome", data=df,
              palette=["skyblue", "salmon"])
plt.title("Age Group vs Diabetes Outcome")
plt.xlabel("Age Group")
plt.ylabel("Number of Patients")
plt.legend(["Non-Diabetic", "Diabetic"])
plt.show()
```



Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

➢ The experiment estimated the population mean glucose level (μ) of the patients. Its estimated value (sample mean, $\bar{x}$) is **120.89 mg/dL**, as indicated on the top-left distribution plot.

2. What confidence interval was obtained, and how should it be interpreted?

➢ Based on the sample size ($n = 768$) and distribution, the estimated 95% confidence interval for the mean glucose level is approximately **$[118.63, 123.15]\text{ mg/dL}$**. This means we are 95% confident that the true population mean glucose level lies within this range.

3. What hypotheses were formulated for the selected statistical test?

➢ For comparing diabetic versus non-diabetic outcomes across groups (such as BMI or age), the null hypothesis (H_0) states that there is no difference or association between the parameters. The alternative hypothesis (H_1) states that a significant difference or association exists between the outcome groups.

4. Based on the p-value, what statistical decision was made?

➢ The statistical decision is to reject the null hypothesis (H_0). The distinct visual separations in BMI and age distributions across diabetes outcomes indicate a highly statistically significant effect with a p-value well below the standard threshold ($p < 0.05$).